

Wednesday Warm Up: Unit 5, Modern Medical Physics

Prof. Jordan C. Hanson

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1 Memory Bank

- **Cross section:** Let n be the number density of atoms in a material, and I_0 be the intensity of radiation particles. The remaining intensity after a distance z is

$$I(z) = I_0 \exp(-n\sigma z) \quad (1)$$

where σ is known as the *cross-section* of the radiation on a material. A unit of cross-section is the barn, which is equal to 10^{-24} cm^2 .

- **Number densities:** The number density of lead, for example, is $3.3 \times 10^{22} \text{ cm}^{-3}$.
- **Units of energy deposition:** 1 rad is equivalent to 0.01 J kg^{-1} of radioactive energy deposited in tissue. 1 Gray is 100 rad, or 1 J kg^{-1} .
- **Relative Biological Effectiveness:** RBE is a qualitative factor by which we multiply the rads to characterize how dangerous or effective radiation is in human tissue. $1 \text{ rem} = 1 \text{ rad} \times \text{RBE}$. $100 \text{ rem} = 1 \text{ Gray} \times \text{RBE} = 1 \text{ Sievert}$, or 1 Sv.
- **Half-life:** Suppose N_0 radioactive isotopes exist as a sample. The time-dependence is

$$N(t) = N_0 \exp(-\lambda t) \quad (2)$$

where $t_{1/2} = \ln(2)/\lambda$ is known as the *half-life*.

2 Propagation of Radiation

1. Suppose x-rays have a cross section of 200 barns at 10 keV in a lead vest protecting a medical patient. If the vest is 1 cm thick, what fraction of the x-rays pass through it?

3. Estimate the number density of human tissue. Suppose, for example, humans are mostly water.

3 Biological Effects of Ionizing Radiation

1. (a) What is the radioactive dose in rads if 250 mJ of total energy is deposited in the body of a 60 kg adult? (b) What instead would be the dose if the radiation was concentrated in just 2.0 kg of tissue? (c) Assuming the RBE is 1 (x-rays or γ rays), what is the dose from part (b) in Sv? (d) Does this dose represent a serious health risk? (Look up the answer).

2. At what thickness would half of the x-rays survive?